Advanced Databases

Learning Objectives *13 August 2026*

- **select an appropriate database technology for a specific use case**
- **design intermediate to advanced relational databases**
- **describe the mechanics of graph databases and other NoSQL technologies**
- **differentiate the affordances of different database technologies**
- **query a real-world data model**

Summary

“This course will explore intermediate-level design and implementation of database systems with an emphasis on scalable, distributed systems. The course will deepen students’ knowledge of advanced relational database management, followed by discussing current and emerging practices for dealing with big data and large-scale database systems used by many social networking and e-commerce services. Concepts include the design and implementation of relational databases, exploration of distributed data structures including graph, document, and key-value storage models, and scalable and resilient query processing. Students will gain practice working with real data and multiple modern database technologies.”

Course Designer

Emily Pfaff	• Assistant Professor of Medicine at the UNC School of Medicine • Co-Director of Informatics and Data Science at NC TraCS • Domain Scientist at the Renaissance Computing Institute

Course Instructor

Adam Lee	• adam_lee@med.unc.edu

	• Research Informatics Specialist at the UNC School of Medicine's Translational and Clinical Sciences Institute • doctorate in Health Informatics from the University of North Carolina at Chapel Hill • Master of Business Administration • Bachelor's degree in information technology